Lecture 3 . Flow Matching

Idea. transport the probability mass from initial to target

$$\int p_{initial}(x)\,dx = 1 \qquad\qquad \int p_{target}(x)\,dx = 1$$

initial dist. target dist.

$p_0 = N(0,1)$ $\qquad$ $p_1 = p_{data}$

$t = 0$ $\qquad$ $t = 1$.

Flow: $\psi_t(x_0)$: collection of trajectories x_t which start from different x_0.

$$\psi : (t, x_0) \longrightarrow \psi_t(x_0).$$

$t \in [0,1)$, $x_0 \in \mathbb{R}^d$, $\psi_t(x_0) \in \mathbb{R}^d$

Probability Path: distribution of x_t at time t. p_t.

Vector Field: (Velocity) $u_t(x)$: where to move (direction, speed) at time t and location x.

$u_t(x) \longrightarrow$

Relationship : Velocity: where to move at each timestep for each instance

Score : compass. (general direction)

Perspective of a single sample with ODE.

$$dx = u_t(x)\, dt.$$

- trajectory starting from x_0 is unique if velocity $u_t(x)$ is Lipschitz continuous.

$\psi_t(x_0)$

Lipschitz continuous: $\exists M, \forall x, y \quad \| f(x) - f(y) \| \leq M \| x - y \|.$

- not varying in a very dramatic way

perspective of a distribution via "mass conservation"

$$\boxed{\text{Temporal evolution of density at time } t \text{ at a particular location}} = \boxed{\text{inflow of density}} - \boxed{\text{outflow of density}}$$

Divergence in 1D:

Divergence ≥ 0

$\frac{\partial f}{\partial x} \geq 0 \qquad \rightarrow \bigcirc \longrightarrow$

$\frac{\partial f}{\partial x} \geq 0 \qquad \leftarrow \bigcirc \longrightarrow$

more leaving than coming

Divergence ≤ 0

$\frac{\partial f}{\partial x} \leq 0 \qquad \leftarrow \bigcirc \longleftarrow$

$\frac{\partial f}{\partial x} \leq 0 \qquad \rightarrow \bigcirc \leftarrow$

more coming than leaving.

Generalization: $\operatorname{div}(f) = \nabla \cdot f = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}$

Divergence of probability flux.

$$\frac{\partial p_t}{\partial t}(x) = -\nabla \cdot (p_t u_t)(x)$$

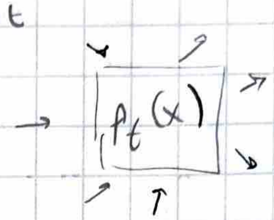
<u>Single Sample</u>

$$\frac{dx}{dt} = u_t(x)$$

$$x \longrightarrow x + dx$$

<u>Distribution</u>

$$\frac{\partial p_t}{\partial t}(x) = -\nabla \cdot (p_t u_t)(x)$$



- vector field $u_t(x)$ generates probability path $p_t(x)$.
- u_t and p_t satisfy the continuity eqn.

<u>single sample</u>

$$x_0 \sim p_0$$

$$\frac{dx_t}{dt} = u_t(x_t)$$

$$\Longrightarrow$$

<u>distribution</u>

$$x_t \sim p_t.$$

Flow Models

Goal. $x_0 \sim p_0$ $\quad x_1 \sim p_1$

Strategy

(1) Training. estimate $u_t(x)$ for all time t and all locations x via $u_t^\theta(x)$.

(2) Inference. sample from initial distribution and solve numerically the ODE using learned vector field $u_t^\theta(x)$

$$\hat{x}_1 = x_0 + \int_0^1 u_t^\theta(x)\, dt.$$

Earlier attempts to learn the velocity

Goal. learn vector field $u_t(x)$ via maximum likelihood

Idea. transform continuity eqn to

$$\frac{d}{dt} \log p_t(x) = - \nabla \cdot u_t(x) .$$

simulate ODE at training time to maximise likelihood

$$\log p_1^\theta(x_1) = \log p_0(x_0) + \int_0^1 - \nabla \cdot u_t^\theta(x)\, dt$$

Limitation. training is slow and expensive

Flow Matching: estimate the vector field with $u_t^\theta(x)$

$$\mathcal{L}_{FM} = \mathbb{E}_{t,x}\Big[\| u_t^\theta(x) - \underbrace{u_t(x)}_{\text{we dont have access to this}} \|^2 \Big] .$$

proposed conditional probability path

$$p_t(x \mid x_1) = \mathcal{N}(t x_1, (1-t)^2 I) .$$

conditional vector field $u_t(\cdot \mid x_1)$ generates conditional probability path $p_t(\cdot \mid x_1)$.

Consequence. if $x_0 \sim p_0(\cdot \mid x_1)$ and $\frac{dx_t}{dt} = u_t(x_t \mid x_1)$

$$x_t \sim p_t(\cdot \mid x_1)$$

$$u_t(x \mid x_1) = \frac{x_1 - x}{1-t}$$

when t close to 1 and x is far from x_1, velocity is high

$$x_t \sim p_t(x_t \mid x_1) = N(t x_1, (1-t)^2 I).$$

$$x_t = t x_1 + (1-t) x_0 \qquad x_0 \sim N(0, 1)$$

$$u_t(x_t \mid x_1) = \frac{x_1 - [t x_1 + (1-t) x_0]}{1-t} = x_1 - x_0.$$

↓

simplifies to this.

$$p_t(x) = \int p_t(x \mid x_1)\, p_{data}(x_1)\, dx_1$$

$$p_0(x) = N(0, I)$$

$$p_1(x) = p_{data}(x).$$

Marginal Vector Field

$$u_t(x) = \int u_t(x \mid x_1) \underbrace{\frac{p_t(x \mid x_1)\, p_{data}(x_1)}{p_t(x)}}_{\text{"}p(x_1 \mid x)\text{"}} dx_1.$$

posterior mean

given where I am, where should I go

- marginal vector field u_t generates marginal probability path p_t.

Deriving Conditional Flow Matching (1.11)

$$\mathcal{L}_{FM} = \mathbb{E}_{t,x}\left[\| u_t^\theta(x) - u_t(x)\|^2\right]$$

equivalent to

$$\mathcal{L}_{CFM} = E_{t,x_1,x}\left[\| u_t^\theta(x) - \underbrace{u_t(x| x_1)}_{x_1 - x_0}\|^2\right].$$

- Same trick for $\|a-b\|^2 = \|a\|^2 - 2\langle a,b\rangle + \|b\|^2$.

derivation

$$\mathop{E}_{t,x_1,x}\left[\langle u_t^\theta(x), u_t(x|x_1)\rangle\right]$$

$$= \int_t\int_{x_1}\int_x \langle u_t^\theta(x), u_t(x|x_1)\rangle\, p_{data}(x_1)\, p_t(x|x_1)\, dx_1\, dx\, dt$$

$$= \int_t\int_x \left\langle u_t^\theta(x), \int_{x_1} u_t(x|x_1)\frac{p_t(x|x_1)\, p_{data}(x_1)}{p_t(x)}\, dx_1\right\rangle p_t(x)\, dt$$

$$= \int_t\int_x \langle u_t^\theta(x), u_t(x)\rangle\, p_t(x)\, dx\, dt.$$

because

$$\int_{x_1} u_t(x|x_1)\frac{p_t(x|x_1)\, p_{data}(x_1)}{p_t(x)}\, dx_1$$

$$= \int_{x_1} u_t(x|x_1)\frac{p_t(x, x_1)}{p_t(x)}\, dx_1$$

$$= \int_{x_1} u_t(x|x_1)\, p(x_1|x)\, dx_1 = \mathop{E}_{x_1\sim p_t(\cdot|x)}\left[u_t(x|x_1)\right] = u_t(x)$$

$$P(A|B) = P(A\cap B)/P(B) = P(B|A)\,P(A)/P(B)$$

Strategy:

1. derive vector field for a simple case (dirac)
2. Construct target vector field for our case (continuity eqn)
3. show loss function is tractable.
4. deduce final loss function ← simple loss

Flow matching for generative modeling

→ normalizing flows, continuous normalizing flows

Training

① $x_0 \sim N(0,1)$, $x_1 \sim p_{data}$, $t \sim \mathcal{U}(0,1)$

noise, clean image, time step

$$x_t = (1-t)\, x_0 + t\, x_1$$ noised image

② use x_t and t to predict $x_1 - x_0$ via $u_t^\theta(x_t)$.

③ Loss / backprop $\mathcal{L} = \| u_t^\theta(x_t) - (x_1 - x_0)^2 \|^2$

backprop through u^θ.

Inference

① $x_0 \sim N(0, I)$ sample

② use Euler to numerically solve ODE

$$x_{t_i} = x_{t_{i-1}} + u_{t_{i-1}}^\theta (x_{t_{i-1}})(t_i - t_{i-1})$$

③ obtain x_1

Discussion

intersecting trajectories

$$\mathcal{L}_{CFM} = \mathbb{E}_{t, x_1, x} \left[\| u_t^\theta(x) - u_t(x \mid x_1) \|^2 \right]$$

averaged trajectories

Fine tuning diffusion models.

step 0. train initial model : 1-rectified flow.

step 1. create paired data out of 1-rectified flow

step 2. train on generated data.

* flows are getting straighter at each fine tune step.